AIM2 Course Project: Midterm Report

Phenotype-Conditioned Drug Repurposing for Undiagnosed Rare Disease Patients via Graph Neural Networks and LLM Hybridization

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Abstract

Rare disease patients frequently lack a confirmed molecular diagnosis, leaving clinical phenotypes as the only structured signal available for guiding treatment. We propose a graph-based model that bypasses the diagnostic bottleneck by taking clinical phenotypes as input and directly ranking ∼7,957 candidate drugs by predicted therapeutic relevance over PrimeKG, without requiring an intermediate diagnosis. Our R-GCN encoder with drug-conditioned cross-attention achieves MRR 0.254 on 108 held-out test diseases, substantially outperforming the structural PageRank baseline (MRR 0.019) and narrowing the gap with TxGNN (MRR 0.399), which requires oracle disease input. A zero-shot GPT-4o baseline (MRR 0.459) outperforms the graph model in aggregate but produces statistically independent rankings (Spearman $p = 0.364$), with complementary strengths on different disease subgroups that naïve score-level fusion fails to exploit. These findings motivate our ongoing work on feature-level graph-LLM fusion and phenotype-anchored GraphRAG to integrate structural and semantic signals without requiring a confirmed diagnosis.

1 Introduction and Project Overview

Rare diseases collectively affect an estimated 400 million people worldwide, yet patients face an average diagnostic odyssey of over 5 years, and an estimated 50% never receive a definitive molecular diagnosis [1, 2, 3]. For these undiagnosed patients, clinical phenotypes are often the only structured clinical signal available. However, existing computational tools tend to address fragments of this problem in isolation: either predicting disease diagnoses from phenotype sets, or recommending drugs given a confirmed disease. Currently, no established system connects phenotype input directly to drug output for patients who may never receive a diagnosis.

To bypass the diagnostic bottleneck and provide actionable therapeutic recommendations for rare disease patients, we propose a graph-based model that takes a set of clinical phenotypes as input and

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directly ranks drugs by predicted therapeutic relevance, without requiring an intermediate diagnosis. We first implemented a transductive R-GCN encoder over PrimeKG with a drug-conditioned cross-attention mechanism that learns to weight phenotypes differently per candidate drug. This model (MRR 0.254) substantially outperformed the structure-only PageRank baseline (MRR 0.019) while narrowing the gap with TxGNN (MRR 0.399), which requires a confirmed diagnosis as input. However, error analysis revealed cold-start drugs and phenotype sparsity as primary failure modes, and a zero-shot GPT-4o evaluation (MRR 0.459) demonstrated that semantic knowledge alone captures drug-disease associations absent from the training graph. These findings motivate two refined directions for our work: (1) a transductive graph model with cross-attention as the core architecture, and (2) a graph-LLM hybrid that combines structural reasoning with semantic knowledge. We hypothesize that these end-to-end approaches can match diagnosis-dependent models and produce trustworthy therapeutic suggestions.

2 Background

2.1 Gaps in Current Methodologies.

A fundamental gap in the current methodology is the disconnect between diagnosis and treatment recommendation. Models are typically optimized for one task or the other, but not both. On the diagnostic side, methods such as SHEPHERD [4] and Phen2Gene [5] predict rare disease diagnoses from patient phenotype sets, but they stop at the diagnosis stage without providing trustworthy treatment recommendations. On the therapeutic side, TxGNN [6] achieves zero-shot drug repurposing over PrimeKG that outperforms prior methods, but requires a confirmed disease node as input, which does not hold for many rare disease patients. A natural alternative might be to cascade the phenotype $\rightarrow$ disease and disease $\rightarrow$ drug stages. However, such two-stage pipelines are vulnerable to uncertainty propagation: when the intermediate disease prediction is ambiguous or incorrect, that error is passed to downstream predictions. No existing end-to-end architecture directly maps clinical phenotypes to drug candidates without an intermediate diagnostic step. Notably, phenotype-driven drug repurposing has shown preclinical promise: Ambesi-Impiombato et al. [7] identified therapeutic candidates for Huntington's disease by matching drug-induced phenotypic signatures against disease model phenotypes without requiring mechanistic understanding, suggesting that phenotype sets alone carry sufficient signal to inform treatment selection.

2.2 Knowledge Graphs for Drug Repurposing.

Biomedical knowledge graphs encode rich relational structure among diseases, drugs, phenotypes, genes, proteins, and pathways, and therefore provide a natural substrate for computational drug repurposing. PrimeKG integrates these heterogeneous biomedical relations into a single graph and has supported strong disease-to-drug prediction results [8, 6]. Graph neural networks such as R-GCN [9], HGT [10], and HAN [11] have shown strong performance on link prediction in heterogeneous graphs because they propagate information across multiple node and edge types while learning relation-specific patterns. These properties make them well suited for our setting, where phenotype signals may connect to drugs only indirectly through shared disease modules, molecular pathways, or protein interactions. At the same time, our results with Personalized PageRank suggest that graph topology alone is not sufficient, motivating a learned model that assigns task-specific weights to different relations and phenotype signals.

2.3 The Emerging Role of LLMs.

Large language models trained on scientific literature encode broad pharmacological and mechanistic knowledge that complements the structured relational information in knowledge graphs. Hybrid architectures that initialize graph node features with language model embeddings of drug mechanisms, protein functional annotations, or phenotype descriptions can provide semantically rich representations even for nodes with sparse graph connectivity [12, 13]. Similarly, molecular structure encoders and protein language models such as ESM-2 [14] offer feature-level representations of drugs and targets that go beyond topological graph signals. These complementary capabilities motivate a modular design philosophy: an end-to-end phenotype-to-drug architecture should be extensible to incorporate molecular, proteomic, and language-derived features as they become available, rather than relying solely on graph topology.

3 Data

Our model is built on PrimeKG, a precision medicine knowledge graph integrating 20 biomedical databases with nodes spanning 17,080 diseases, 7,957 drugs, 15,311 phenotypes, 29,786 proteins, and additional biological entities across 10 node types and 30 relation types. PrimeKG contains 300,634 `disease_phenotype_positive` edges and 18,776 indication edges, which together support phenotype-to-drug learning using supervision contained within the graph itself. We additionally leverage 5,136 off-label use edges, which connect drugs prescribed for diseases outside their approved indications, as auxiliary positive supervision. This refinement was not part of our original proposal but was introduced after ablation experiments showed improved performance.

The input to our model is a set of phenotype node indices for a given disease, and the target outcome is a ranked list of all candidate drugs scored by predicted therapeutic relevance, evaluated against known indication and off-label edges as ground-truth labels. We identify 539 diseases that simultaneously have at least one phenotype association and one drug indication, covering 3,518 phenotypes and 1,352 unique drugs. We apply an 80/20 disease-level split that yields 431 training diseases (2,703 disease-drug pairs) and 108 test diseases (784 pairs). Each test disease is represented by an average of 23 phenotypes that serve as the sole input at inference. To prevent information leakage through message passing, all edges incident to test disease nodes are removed from the training subgraph. Of the 485 unique drugs in the test set, 345 (71.13%) also appear in training indications, indicating a potential popularity bias toward frequently indicated drugs.

4 Baseline Implementation

4.1 Personalized PageRank

Our structural baseline requires no learned parameters and tests whether graph topology alone carries sufficient signal for phenotype-to-drug prediction. We constructed an undirected, row-normalized adjacency matrix over all 129,375 PrimeKG nodes, excluding all edges incident to test disease nodes to match the information constraints of the learned model. For each test disease, the associated phenotype nodes served as seed nodes for Personalized PageRank (PPR) with restart probability $\alpha = 0.15$. We computed PPR scores via power iteration (up to 50 iterations, convergence threshold 10^{-6}): at each step, the score vector is updated as

$$\mathbf{r} \leftarrow (1-\alpha)\, A^{\top}\mathbf{r} + \alpha\, \mathbf{s},$$

where $\mathbf{s}$ is the uniform seed vector over phenotype nodes. After convergence, all 7,957 drug nodes are ranked by their PPR scores in descending order. We additionally conducted a sensitivity analysis over $\alpha \in \{0.05, 0.10, 0.15, 0.20, 0.30, 0.50, 0.75\}$ on a 30-disease subset and found stable performance across all values.

4.2 TxGNN (Upper Bound)

We use TxGNN as an upper-bound baseline representing the best achievable performance when a confirmed disease diagnosis is available. TxGNN is a relational graph convolutional network trained on PrimeKG for drug-disease link prediction, combined with a disease similarity-based metric learning module that enables zero-shot prediction for diseases with no known treatments. To evaluate TxGNN on our split, we removed all disease-drug indication and reverse-indication edges for the 108 test diseases from the training graph (1,506 edges total) and retrained from scratch using the official TxGNN package. The test disease nodes and all their non-indication edges remain in the graph, giving TxGNN strictly more information than our model and justifying its role as an upper bound.

Node embeddings of dimension 512 are learned through a two-layer R-GCN encoder over PrimeKG's heterogeneous graph. Training proceeds in two phases: a pretraining phase (2 epochs, learning rate 10^{-3}, batch size 1,024) performing all-relation link prediction across the full graph to learn general node representations, followed by a fine-tuning phase (500 epochs, learning rate 5×10^{-4}) optimizing drug-disease indication prediction using binary cross-entropy loss with the AdamW optimizer and ReduceLROnPlateau scheduler. Model selection is based on the best validation macro AUROC on the edge level.

4.3 Phenotype Similarity + TxGNN Cascade

Our third baseline is a two-stage, phenotype $\rightarrow$ disease $\rightarrow$ drug cascading pipeline. The cascade follows the same conceptual pipeline as chaining SHEPHERD with TxGNN, but replaces SHEPHERD's learned GNN diagnosis model with parameter-free Jaccard retrieval over phenotype profiles [15, 16]. We adopt this simplification for two reasons: SHEPHERD's codebase requires legacy dependencies incompatible with TxGNN's runtime environment, and a non-learned diagnostic step isolates the effect of the cascade architecture from confounding differences in model capacity.

The cascade operates in three steps. First, for each test disease's phenotype set, we compute Jaccard similarity against all 431 training diseases' phenotype profiles and retrieve the top-K most similar training diseases. Second, we query the retrained TxGNN model to score all candidate drugs for each of the K retrieved diseases. Third, we aggregate drug scores as a weighted average:

$$\text{score}(\text{drug}) = \frac{\sum_{i=1}^{K} \text{sim}_i \cdot \text{TxGNN}_i}{\sum_{i=1}^{K} \text{sim}_i},$$

where sim_i is the Jaccard similarity of the i-th retrieved disease. The cascade requires no additional training and has one hyperparameter K. We evaluated $K \in \{1, 3, 5, 10, 20\}$ and report the best configuration ($K = 1$).

4.4 GPT-4o Zero-Shot LLM Baseline

Our fourth baseline tests whether a large language model can predict therapeutically relevant drugs from phenotype descriptions alone, without using graph structure [17, 18]. For each of the 108 test diseases, we prompted GPT-4o with the disease's HPO phenotype names and asked it to return the 50 most therapeutically relevant drugs in ranked order as a JSON array of generic drug names, using temperature 0 for deterministic output. The prompt followed Yan et al. [17]:

> *"A patient presents with the following clinical phenotypes: {phenotype list}. Based on these symptoms, suggest the 50 most therapeutically relevant drugs that could treat the underlying condition(s). Return ONLY a JSON array of drug names, ranked from most to least relevant. Use generic drug names."*

On average, 49 out of 50 suggested drug names per query were successfully identified as existing drugs in PrimeKG through exact and fuzzy string matching. AUROC and AUPRC are not reported for GPT-4o because the model generates a fixed-length ranked list rather than continuous scores over all candidate drugs.

4.5 Baseline Results

We evaluated all baselines using the following ranking and classification metrics:

- **Mean Reciprocal Rank (MRR).** For each test disease, we compute the reciprocal rank of the highest-ranked true drug, and then average this quantity across all test diseases. A higher MRR indicates that at least one correct drug is ranked closer to the top of the list.
- **Recall@K.** For each test disease, Recall@K is the fraction of true drugs that appear among the top-K ranked predictions. We report the average Recall@K across all test diseases. This metric reflects whether a clinician reviewing the top-K recommendations would encounter a true drug.
- **AUROC (Area Under the Receiver Operating Characteristic Curve).** For each test disease, AUROC measures the probability that the model assigns a higher score to a randomly chosen true drug than to a randomly chosen non-indicated drug. We report the average AUROC across all test diseases.
- **AUPRC (Area Under the Precision–Recall Curve).** For each test disease, AUPRC summarizes the precision–recall trade-off over all possible score thresholds, and we average this value across all test diseases. AUPRC is particularly appropriate in our setting because fewer than 0.1% of drug–disease pairs are positive indications.

Table 1: Baseline drug ranking performance measured by Mean Reciprocal Rank (MRR), Recall@K (K = 1, 5, 10, 50), AUROC, and AUPRC on 108 held-out test diseases.

Model	MRR	R@1	R@5	R@10	R@50	AUROC	AUPRC
PPR	0.019	0.000	0.000	0.033	0.088	0.811	0.790
Cascade	0.121	0.042	0.095	0.142	0.233	0.856	0.820
GPT-4o	0.459	0.113	0.239	0.308	0.456	—	—
TxGNN	0.399	0.141	0.303	0.390	0.749	0.945	0.923

Table 1 summarizes ranking performance across four baselines. PPR establishes a performance floor (MRR 0.019, near-zero Recall@1 and Recall@5), confirming that graph structure alone carries little useful signal for phenotype-to-drug prediction. TxGNN performs substantially better (MRR 0.399, Recall@50 0.749), reflecting the advantage of oracle disease input. The cascading pipeline improves over PPR (MRR 0.121) by leveraging TxGNN's learned scores from phenotypically similar training diseases, but the two-stage design still loses substantial information. GPT-4o achieves the highest MRR (0.459), demonstrating that a large language model can place a relevant drug very high in the ranking even without explicit graph structure.

For Recall@K, TxGNN performs best overall, with Recall@1 of 0.141, Recall@5 of 0.303, Recall@10 of 0.390, and Recall@50 of 0.749. GPT-4o shows a different pattern. Its MRR is the highest, but its recall is lower than TxGNN at every K. This pattern suggests that GPT-4o is often able to rank one relevant drug near the top, but it retrieves fewer relevant drugs overall. The cascade remains well below TxGNN across all recall metrics. This result confirms that replacing the true disease with phenotypically similar diseases weakens downstream drug ranking. Overall, these findings show that oracle disease input remains highly valuable for broad drug retrieval, while GPT-4o provides a strong phenotype-only baseline for early-rank precision. Figure 1 provides a visual comparison of both MRR and Recall@K.

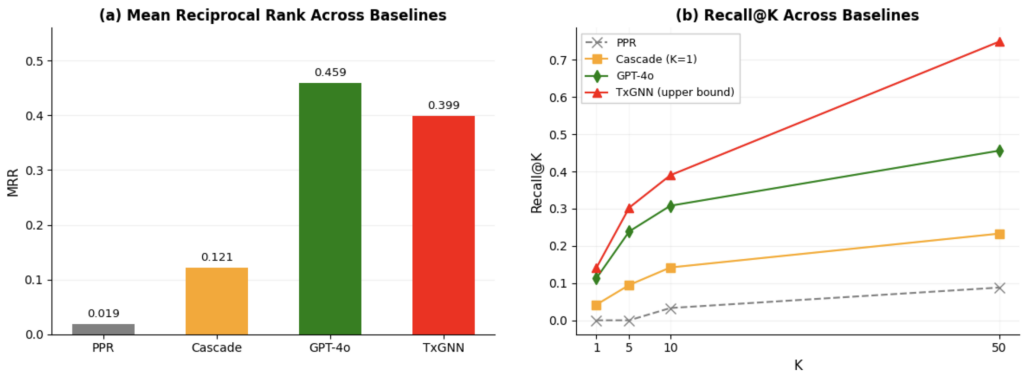

Figure 1: Visual comparison of baseline ranking performance. Panel (a) shows mean reciprocal rank (MRR) across methods. Panel (b) shows Recall@K at $K = 1, 5, 10$, and 50.

AUROC and AUPRC reveal new patterns. PPR achieves AUROC 0.811 despite near-zero MRR, and the cascade shows a similar gap (AUROC 0.856, MRR 0.121). In both cases, the model assigns higher scores to true drugs than most non-drugs but fails to rank them above a small number of high-scoring popular false positives. TxGNN achieves the highest AUROC (0.945) and AUPRC (0.923), which are consistent with its strong ranking performance. This gap between classification and ranking metrics across methods confirms popularity bias: even when a model scores true drugs higher than most non-drugs (high AUROC), broadly indicated drugs consistently occupy top positions in the ranked list and push the correct but less common treatments further down (low MRR).

5 Proposed Method: Design and Implementation Status

5.1 Implemented and Tested: R-GCN with Drug-Conditioned Cross-Attention

Our model consists of two components trained end-to-end. The **encoder** is a 2-layer Relational Graph Convolutional Network (R-GCN) with basis decomposition (10 bases) operating over PrimeKG's masked training subgraph ($\sim$8M edges across 60 relation types including reverse edges). It produces 128-dimensional context-aware embeddings for all 129,375 biomedical entities by aggregating multi-hop neighborhood information through relation-specific message passing. The **scorer** is a drug-conditioned cross-attention mechanism with 4 attention heads: for each candidate drug, its embedding serves as the query while the phenotype embeddings serve as keys and values. This allows the model to weight phenotypes differently per drug—for example, emphasizing cardiovascular phenotypes when scoring an antihypertensive—rather than collapsing the phenotype set through mean pooling. The attended output is projected to a scalar relevance score.

The model is trained with a margin-based ranking loss (margin $= 1.0$). For each (phenotype set, true drug) pair derived from indication and off-label use edges, one negative drug is sampled using degree-weighted sampling, where drugs with more training indications are sampled more frequently as negatives to counteract popularity bias. Training uses Adam (LR $= 10^{-3}$, weight decay $= 10^{-5}$) with gradient accumulation over 4 steps and gradient clipping at 1.0. All test disease nodes and their incident edges are fully removed from the training graph prior to model training, ensuring zero information leakage. At inference, the model scores all 7,957 PrimeKG drugs in a single forward pass given only a held-out phenotype set, without ever constructing a test disease node embedding. The model achieves MRR 0.254 on the 108 held-out test diseases.

We report results using test-set model selection rather than a held-out validation split. Partitioning 10% of training diseases as a validation set (43 diseases) reduced training samples from 7,000 to 3,000 and dropped test MRR to 0.11, as the validation MRR over only 43 diseases was too noisy to reliably select checkpoints. Given the limited scale of our dataset, we instead use early stopping based on loss convergence and note that test-set selection may yield mildly optimistic estimates.

Ablation Experiments. We conducted eight ablation experiments, each modifying one aspect of the base configuration: (1) adding off-label use edges as training positives; (2) ensembling model scores with PPR at inference; (3) adding residual connections in the encoder; (4) lowering learning rate and margin; (5) sampling 30% of negatives from contraindication edges; (6) increasing the negative ratio to 3:1; (7) replacing margin loss with InfoNCE loss (temperature $= 0.1$, 5 negatives); and (8) end-to-end pretraining with DistMult link prediction followed by fine-tuning with validation-based model selection. Of these, only off-label positives and PPR ensembling improved performance; all other modifications degraded MRR, consistent with the small training set ($\sim$7,000 pairs across 431 diseases) being the binding constraint.

Error Analysis. We identified three systematic failure modes: (1) **Cold-start drugs** that never appear in any training indication or off-label edge (8/108 test diseases, MRR ≈ 0); (2) **Phenotype sparsity**, where diseases with 1–3 phenotypes achieve substantially lower MRR than those with 11+ phenotypes (MRR 0.13 vs. 0.31); and (3) **Popularity bias**, where high-degree drugs such as Stanolone and Ribavirin are systematically over-ranked across test diseases.

Comparison with LLM. We evaluated the R-GCN model against the GPT-4o baseline on 108 held-out test diseases. The LLM substantially outperformed the R-GCN in aggregate (MRR 0.459 vs. 0.254), driven primarily by pretraining on biomedical literature providing a direct recall advantage for well-characterized diseases. However, a Spearman rank correlation test between the two models' rankings yielded $p = 0.364$, confirming that the R-GCN and LLM produce statistically independent, non-redundant signals.

Quadrant analysis highlights that the R-GCN functions as a "rare disease specialist," outperforming the LLM on 36 diseases—specifically those characterized by higher phenotype density (25.6 avg. phenotypes) and sparse therapeutic landscapes (3.9 avg. true drugs), where the LLM's semantic memorization fails. For niche conditions such as Wolman disease and Glanzmann thrombasthenia, the model achieved MRR 1.000 against the LLM's 0.000, demonstrating that graph-based structural reasoning captures drug–phenotype associations that semantic knowledge alone overlooks. Conversely, both methods fail together on poorly characterized diseases with sparse annotations ($n = 25$,

avg. 2.4 true drugs), and the LLM dominates on common diseases where drug associations are well-documented in training corpora.

Despite this complementarity, naïve fusion strategies failed to capture meaningful synergy. Reciprocal rank fusion (RRF), linear score interpolation with z-score normalization, and min-max normalization across mixing weights $\beta \in \{0.1, \ldots, 0.9\}$ all degraded MRR below the LLM alone. The sole strategy that improved performance was freezing the LLM's top-10 ranked drugs and using the R-GCN to re-rank only the remaining candidates (MRR 0.4718 vs. 0.4658). This marginal gain from heuristic partitioning, together with the failure of all parametric fusion strategies, motivates the transition from score-level fusion to degree-conditioned feature-level fusion described in Section 5.2.

5.2 Planned Experiments Beyond the Baseline

Two directions remain under active development, motivated by the complementary failure modes above.

5.2.1 LLM-Enhanced Fusion Approaches

Method 1: Late Fusion. Our R-GCN and zero-shot LLM baseline exhibit complementary failure modes, where the graph model fails on cold-start drugs while the LLM lacks structural grounding, suggesting that combining their scores at inference may yield additive gains. Without modifying either model, we propose to combine their ranking scores at inference:

$$s_{\text{final}}(d) = \beta \cdot s_{\text{graph}}(d) + (1 - \beta) \cdot s_{\text{LLM}}(d),$$

where $s_{\text{graph}}(d)$ is the scalar output of the trained R-GCN cross-attention scorer and $s_{\text{LLM}}(d)$ is the cosine similarity between the mean-pooled text embedding of the patient's phenotype set and the text embedding of drug d. All entity embeddings are pre-computed and cached, so inference requires only a dot-product lookup. β is tuned on validation MRR. This approach requires no retraining and directly tests whether the two models' errors are complementary.

Method 2: Feature-Level Fusion. Rather than combining scores, we integrate LLM-derived semantic embeddings directly into the R-GCN's representation space. We generate textual descriptions for all 7,957 drugs and 15,311 phenotypes seeded with existing PrimeKG metadata, and compare four text encoders: PubMedBERT [19] as a baseline; PubMedBERT-large/BiomedBERT as a scaled successor [20]; BioLinkBERT [21], which improves multi-hop biomedical reasoning through citation-linked pretraining; and SPECTER2 [22], optimized for scientific abstract-level semantic similarity. All encoder outputs are projected to 128 dimensions and L2-normalized before fusion.

Approach (a) — Degree-Conditioned Weighted Averaging. The fused embedding is:

$$\mathbf{h}_{\text{fused}} = \alpha(v)\, \mathbf{h}_{\text{graph}} + (1 - \alpha(v))\, \mathbf{h}_{\text{LLM}},$$

where $\alpha(v) = \sigma(w \cdot \log(\deg(v) + 1) + b)$ with w, b shared learned scalars. Conditioning on degree rather than the graph embedding is deliberate: cold-start node embeddings carry no reliable signal, while degree is always available from graph structure. Gate parameters are frozen before ranking fine-tuning to prevent absorption of popularity bias. A semantic anchoring regularizer

$$\mathcal{L}_{\text{anc}} = \|\mathbf{h}_{\text{fused}} - \mathbf{h}_{\text{LLM}}\|^2$$

prevents graph gradients from overwriting LLM signal for sparse nodes, following LLaDR [23].

Approach (b) — Autoencoder Fusion. Following LLM-DDA [24], which demonstrated $+23.2\%$ AUPR over concatenation, we train a shallow autoencoder in two phases. Unsupervised pretraining over all 129,375 PrimeKG nodes via reconstruction loss produces meaningful fused embeddings for cold-start drugs without requiring indication labels. Supervised fine-tuning then optimizes the ranking loss using the latent code in place of node embeddings in the cross-attention scorer.

For clinical deployment, we additionally consider an agentic fallback querying DrugBank [25] and ClinicalTrials.gov [26] for drugs scoring below a confidence threshold. This is excluded from quantitative evaluation due to reproducibility constraints and potential leakage from live trial registries, but is envisioned as a post-hoc augmentation presenting retrieved evidence to clinicians with explicit provenance.

5.2.2 Phenotype-Anchored GraphRAG

Existing GraphRAG systems (KG-RAG [27], CLADD [28], ESCARGOT [29]) anchor retrieval to a confirmed disease node, which is unavailable in our undiagnosed setting. We instead anchor directly to the patient's HPO phenotype nodes and traverse two disease-free path types in parallel across PrimeKG:

- **Path Type 1: Direct Target:** phenotype → protein → drug via `gene_phenotype` and `drug_protein` edges, capturing drugs that directly modulate phenotype-associated proteins.
- **Path Type 2: Pathway-Mediated:** phenotype → protein → pathway → drug via `protein_pathway` and `drug_pathway` Reactome edges, capturing drugs acting on the same biological pathway even without a direct protein-drug edge.

The union of both path types forms the candidate pool $\mathcal{D}$. Following K-Paths [30], which demonstrated 90% graph size reduction with maintained performance in cold-start settings, retrieved paths are pruned by jointly maximizing relevance (cosine similarity between serialized path embeddings and the mean-pooled phenotype set description) and diversity, greedily selecting paths covering distinct proteins and relation types rather than redundant routes through high-degree hub proteins.

Pruned paths are serialized as structured natural language using entity descriptions cached from the LLM fusion pipeline, and each drug in $\mathcal{D}$ is annotated with its phenotype coverage count across both path types, providing the LLM a structural prior reflecting clinical intuition that drugs addressing multiple phenotypic dimensions are more therapeutically compelling. Following Sanz et al. [31], we apply chain-of-thought prompting to GPT-4o to produce a final ranked drug list with explicit reasoning traces. For drugs not retrieved into $\mathcal{D}$, ranking scores default to the text embedding cosine similarity from Section 5.2 to ensure all 7,957 candidate drugs receive a score at evaluation time.

6 Challenges, Risks, and Mitigation

6.1 Data Constraints

While PrimeKG is of a large scale, our disease-level focus results in relatively small train/test sets from a 539-disease subset, and hence limited supervision. With only 2,703 positive disease-drug pairs across 431 training diseases, the effective number of gradient signals per drug is small, and ablation experiments confirm that most architectural modifications degrade rather than improve performance. Our mitigation has been to retain the simplest effective configuration and pursue complementary signals rather than increasing model complexity.

Our second concern is that our ground truth is limited to known indication and off-label edges in PrimeKG, which is comprehensive but still incomplete. A model that correctly identifies a clinically plausible repurposing candidate absent from PrimeKG would be penalized as a false positive. Our plan to incorporate LLM features may help address this issue during training, since semantic embeddings can provide biomedical knowledge that is not explicitly encoded in PrimeKG. For fairness during evaluation, we plan to evaluate the final model against external validation sources to assess whether top-ranked candidates generalize beyond PrimeKG's known edges.

6.2 Cold-Start and Sparse-Support Diseases

Our error analysis revealed that 8 out of 108 test diseases are fully cold-start, with none of their true drugs appearing in any training disease-drug pair. Both the graph model (MRR ≈ 0 for cold-start diseases) and GPT-4o (1 out of 8 cold-start drugs found) fail in these cases, which confirms that data scarcity for rare disease treatments is a shared fundamental limitation rather than an architecture-specific weakness. More broadly, diseases with sparse phenotype profiles showed worse ranking performance across all methods. We have tried solving this by adding off-label use edges as auxiliary positive supervision, which improved MRR from 0.20 to 0.25 in ablation. GPT-4o also demonstrated that external biomedical knowledge can recover some signals missing from the graph. In the final stage of the project, we plan to use these findings to guide graph-LLM fusion experiments, where LLM-derived drug embeddings may provide meaningful representations for drugs that lack training indication edges.

6.3 Difficulty of Integrating Graph and LLM Signals

Our results show that graph structure and LLM knowledge capture complementary signals: GPT-4o excels on well-documented diseases with many known drug associations, while the R-GCN performs better on diseases with richer phenotype profiles and fewer known treatments. However, combining these signals is nontrivial. Reciprocal rank fusion and linear score interpolation both failed to outperform the LLM alone, despite low correlation between the two models' rankings. This indicates that the difficulty lies not only in identifying complementary signals, but also in reconciling two models with very different score distributions and error patterns, as GPT-4o produces a fixed-length ranked list of drug names, while the R-GCN outputs continuous scores over all 7,957 candidate drugs. To address this, we moved away from score-level fusion. Our next step is feature-level fusion, where frozen biomedical text embeddings for drugs and phenotypes serve as fixed-dimensional input features for the graph model. This design allows semantic knowledge to enter the model during representation learning rather than only after ranking.

6.4 Compute Constraints

All experiments are conducted on Google Colab Pro (single GPU, $\sim$40 GB VRAM). Full-graph training over PrimeKG's 129,375 nodes and $\sim$8M edges is feasible under this budget for the R-GCN, but scaling to inductive architectures or large LLM encoders requires careful memory management. We mitigate this by caching all text embeddings offline and using gradient accumulation to simulate larger effective batch sizes. GraphRAG subgraph extraction is performed CPU-side to avoid GPU memory pressure during LLM inference.

7 Updated Timeline and Plan to Final Report

- **Week 9:** Generate and cache textual descriptions for all drugs and phenotypes using PrimeKG metadata. Encode with each of the four candidate text encoders (PubMedBERT, BiomedBERT, BioLinkBERT, SPECTER2) and evaluate via the late fusion baseline to select the best encoder. Implement and evaluate inference-time degree debiasing in parallel.
- **Week 10:** Implement both feature-level fusion approaches (degree-conditioned weighted averaging and autoencoder fusion) and run a controlled comparison across candidate text encoders to finalize encoder selection.
- **Week 11:** Build the phenotype-anchored GraphRAG pipeline, including subgraph extraction along both path types, path pruning, path serialization, and chain-of-thought LLM ranking over all 108 test diseases.
- **Week 12:** Consolidate all results into a unified comparison table, prepare figures and stratified error analyses (by phenotype count, drug degree, and disease category), and draft the final report.
- **Week 13:** Finalize the report and prepare the final presentation.

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